Macroeconomic Market Analysis

This report analyses the performance of the US and UK equity markets since 2021, focusing on the impact of inflation, interest rates and broader macroeconomic conditions. Using historical market data and Python-based analysis, the report compares the performance, returns and volatility of the S&P 500 and FTSE 100. While both markets experienced significant volatility following the COVID-19 pandemic, the S&P 500 delivered stronger long-term performance, supported by the growth of large technology companies. The FTSE 100 provided comparatively lower returns but demonstrated greater resilience due to its concentration in energy, financial and defensive sectors.

The global economy has faced several major challenges since 2021. Following the COVID-19 pandemic, demand recovered rapidly while supply chains remained disrupted, contributing to high inflation across many economies. In response, central banks such as the **Federal Reserve** and the **Bank of England** increased interest rates aggressively to reduce inflationary pressures.

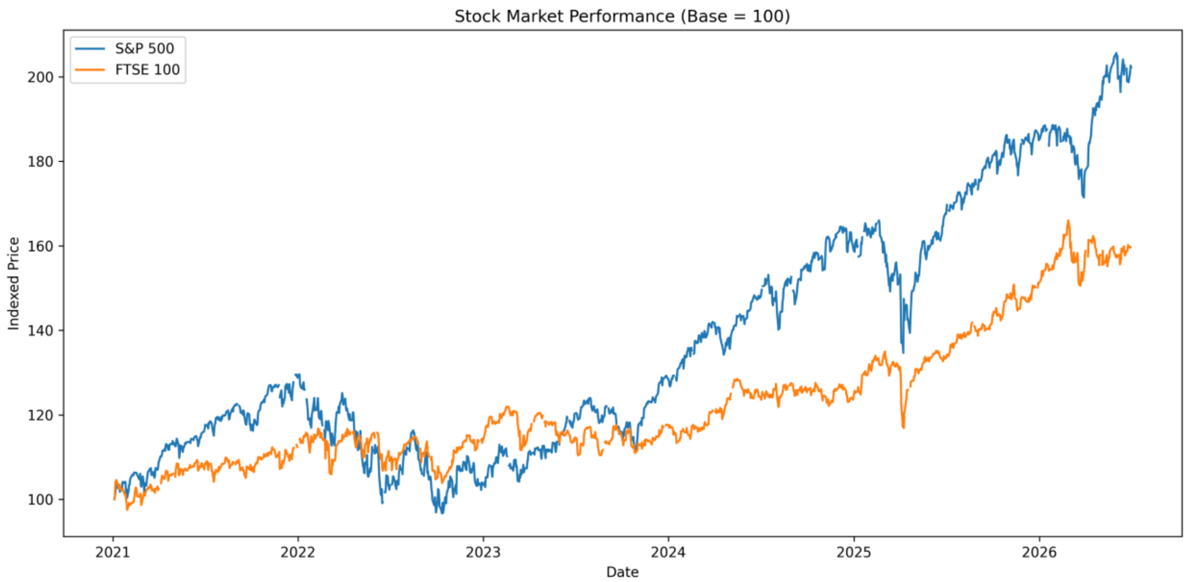
Higher interest rates increase borrowing costs for both businesses and consumers. As a result, economic growth often slows, and investors reassess company valuations, particularly for high-growth technology firms whose future earnings become less valuable when discounted at higher interest rates.

Although inflation has gradually moderated, central banks have maintained relatively restrictive monetary policy to ensure inflation returns to target levels. These economic conditions have continued to influence financial markets.

Python analysis shows that both equity markets recovered from the market volatility experienced during 2022. However, the S&P 500 significantly outperformed the FTSE 100 over the sample period.

The stronger performance of the S&P 500 can largely be attributed to the substantial weighting of technology companies such as Apple, Microsoft and NVIDIA, which benefited from strong earnings growth and increased investment in artificial intelligence.

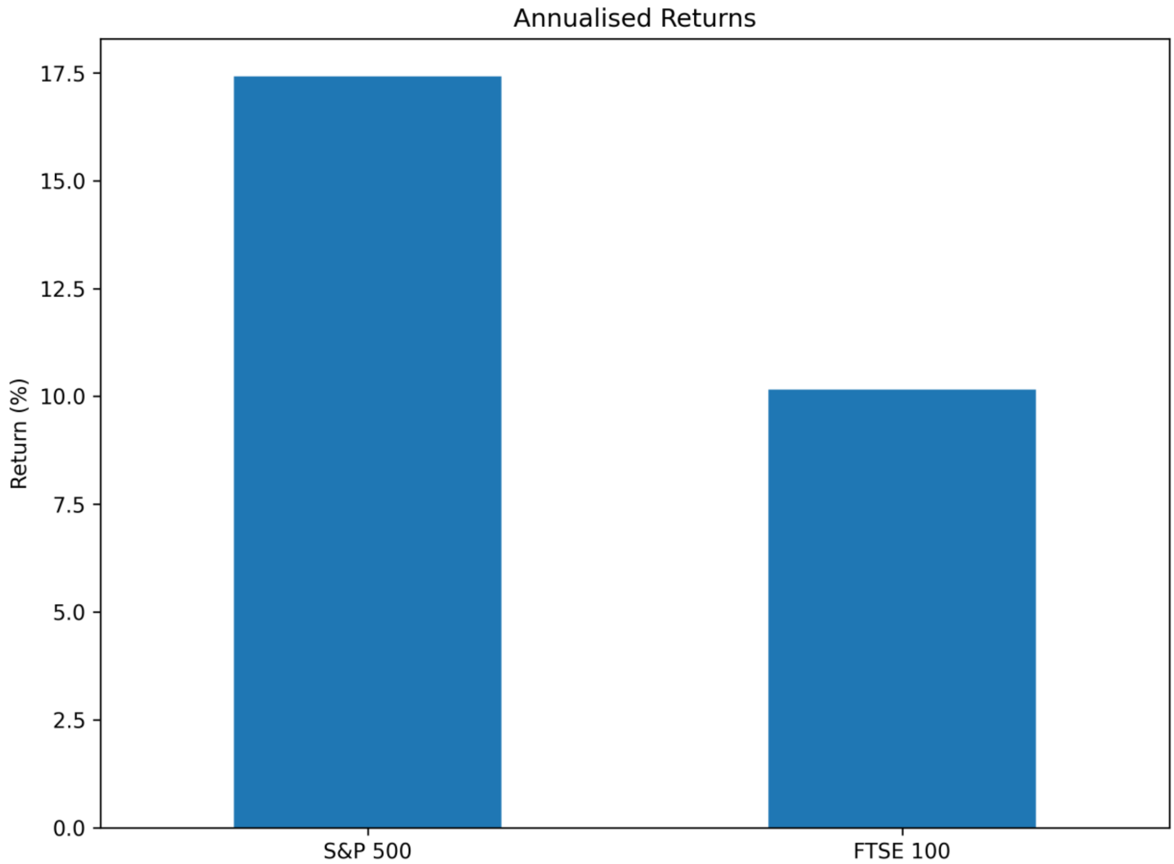
In contrast, the FTSE 100 contains a larger proportion of mature businesses operating within sectors such as banking, energy and mining. While these companies generally produce more stable cash flows, they typically offer lower long-term growth than technology-focused firms.

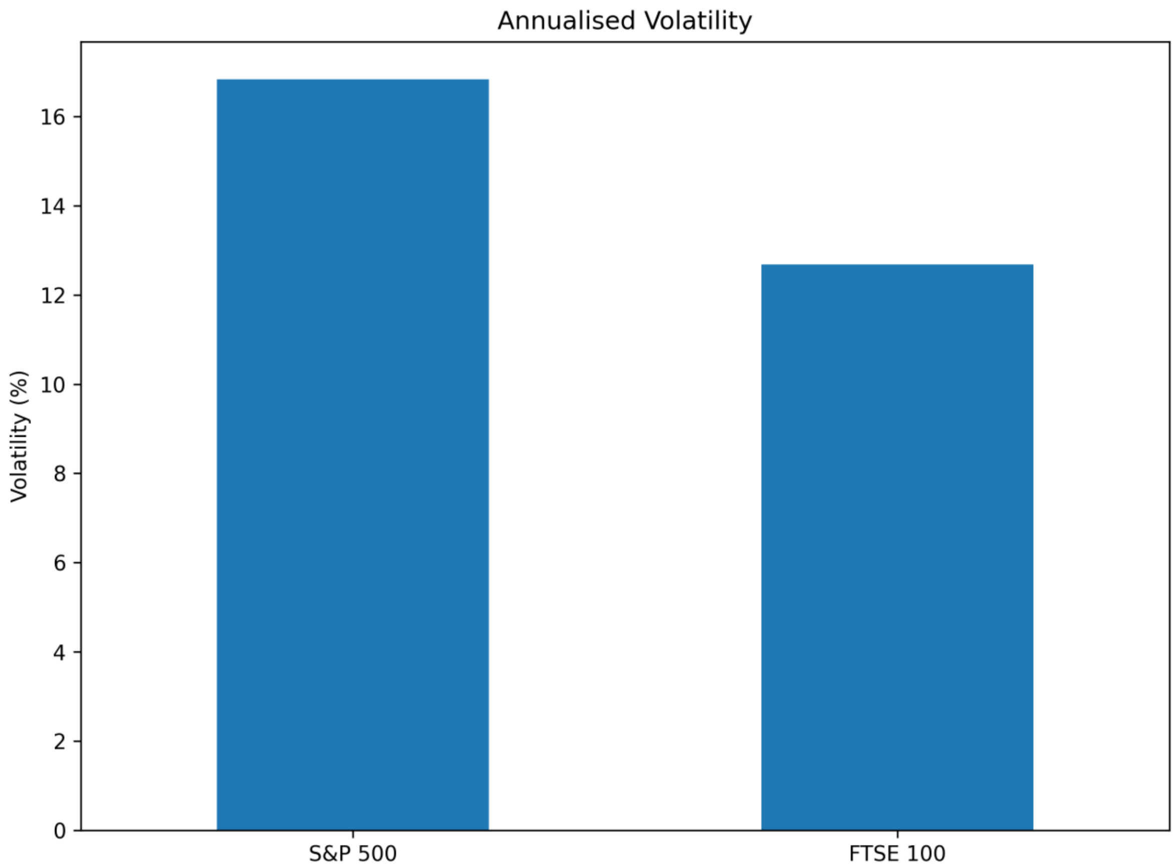


The annualised return calculations indicate that the S&P 500 generated stronger long-term returns than the FTSE 100. This reflects investors' willingness to pay higher valuations for companies with stronger earnings growth.

The volatility analysis demonstrates that although the S&P 500 achieved higher returns, it also experienced greater fluctuations in price. The FTSE 100 exhibited comparatively lower volatility due to its diversified exposure to defensive sectors and dividend-paying companies.

This highlights one of the fundamental principles of investing: higher expected returns are often accompanied by higher levels of risk.


Annualised Returns
Return (%)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
S&P 500
FTSE 100


Annualised Volatility
Volatility (%)
0
2
4
6
8
10
12
14
16
S&P 500
FTSE 100

The correlation matrix indicates that both markets are positively correlated, meaning they generally move in the same direction during periods of economic expansion and contraction. However, the correlation is not perfect, suggesting that international diversification can still reduce overall portfolio risk.

Diversification across different geographic regions and industries remains an important risk management strategy for long-term investors.

Overall, the analysis suggests that the S&P 500 has delivered superior long-term performance compared with the FTSE 100, although this has been accompanied by higher volatility. The FTSE 100 has demonstrated greater stability due to its sector composition but has generated lower returns over the same period.

Looking ahead, future market performance will largely depend on the path of inflation, central bank interest rate decisions and corporate earnings growth. Investors should continue monitoring macroeconomic developments while maintaining diversified portfolios to manage risk effectively.